

ARM-450 — Project Review Dossier

A 450 mm printed 6-DOF manipulator for sinusoidal tracing and probe-and-drogue docking

Prepared for technical inspection · 2026-08-28 · status: pre-hardware

Read this first. Nothing in this project has been printed or measured. Every number is analytical or simulated, and the document says so at each point. That is deliberate: a prediction written down before the test is evidence; the same number written after is not. §10 lists seven measurements not yet made, each with its falsification condition.

1. The idea, and why this design and not another

1.1 The problem

The arm is the manipulation element of MASCOT, a maglev-floated chaser/target capture project. Two demonstrations were required:

1. Sinusoidal trace on a board — a precision demonstration. The traced curve is a permanent, visible record of repeatability, and it exercises five joints at once.
2. Probe-and-drogue docking with fluid transfer — a capture tolerance demonstration.

These fail for opposite reasons, and that shaped everything.

1.2 Why 6-DOF serial, and not the alternatives

architecture	rejected because
SCARA (4-DOF)	cannot orient a tool arbitrarily. Docking needs the probe axis aligned with the port in 3-D; SCARA holds the tool vertical always.
Delta / parallel	excellent stiffness and speed, but a small workspace relative to its footprint, and no natural way to put a rotating tool at the end — the docking latch is driven by tool roll.
Cartesian gantry	the stiffest option and the easiest to build accurately, but it is not a manipulator. It cannot represent a spacecraft arm, which is the point of the project.
5-DOF serial	one short. A 5-DOF arm can position and point, but cannot roll about the tool axis — and tool roll is what latches the docking bayonet. Removing J6 would force a seventh actuator back in elsewhere.

architecture	rejected because
7-DOF redundant	the honest answer for a real space arm (obstacle avoidance, singularity escape). Rejected here on mass: a seventh ST3215 is 60 g plus driver and wiring, against a 1450 g budget already tight.

6-DOF serial is the minimum that does both tasks. That is the argument — not that it is best in the abstract.

1.3 Why these particular design choices

One joint interface on all six axes. Ø42 pocket · 6806 bearing · Ø30 shaft, everywhere. The alternative — sizing each joint to its own load — is what an engineer's instinct says. It was rejected because a printed bearing pocket must be characterised on the machine that prints it, and six different pockets means six characterisations. With one, a single 0.9 h test coupon gates the entire 21 h print. The cost is a wrist carrying more bearing than it needs, and that cost was measured, not assumed: swapping the wrist to a 4 mm-wide 6706 saved 24 g and cost no stiffness, because moment stiffness goes as bearing spacing squared and is independent of bearing width.

Clamshell links, split on the neutral axis. A closed box section 47.5 × 29.0 mm in a 2.4 mm wall. An open C-section of the same envelope has roughly two orders of magnitude less torsional stiffness, and the payload hangs off-axis. The seam bolts are what close the section — they are structural, not cosmetic.

Two bearings per joint, not one. A single deep-groove bearing is a pivot: it holds the shaft centred and lets the joint tilt. Two spaced apart form a couple that resists bending. An earlier revision of this arm used a single thrust washer per joint and it is recorded as the largest error in the project — it produced the wobble and the limit-cycle oscillation that started the redesign.

tool error at 360 mm reach			
single thrust washer	28.90 mm		
bearing pair, no preload	6.95 mm		
bearing pair, some preload	1.44 mm		
preloaded pair, 40 mm apart	0.45 mm	<-	the design

Docking latched by J6, with no seventh actuator. J6 has ±175° of travel and does nothing during a docking approach — a probe is rotationally symmetric about its own axis. Rolling it 30° turns a three-lug bayonet. The motion was already free.

2. Mathematics

2.1 Kinematic chain

world	→	base_link	→	link1	→	link2	→	link3	→	link4	→	link5	→	link6	→	tcp
		J1 Z(yaw)		J2 Y		J3 Y		J4 Z		J5 Y		J6 Z				
z, mm:		50		40		119		119		62		30		30		

Total 450.00 mm base to tool face, verified to 1×10^{-6} mm against the parameter file.

2.2 Forward kinematics — modified (Craig) DH

Frame i is attached to link i , parameters $(\alpha_{i-1}, a_{i-1}, d_i, \theta_i)$:

$${}^{i-1}_i T = \begin{bmatrix} c\theta & -s\theta & 0 & a \\ s\theta \cdot c\alpha & c\theta \cdot c\alpha & -s\alpha & -s\alpha \cdot d \\ s\theta \cdot s\alpha & c\theta \cdot s\alpha & c\alpha & c\alpha \cdot d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	α_{i-1}	a_{i-1} (mm)	d_i (mm)	θ_i
1	0°	0	90	$q_1 + 180^\circ$
2	90°	0	0	$q_2 + 90^\circ$
3	0°	119	0	$q_3 + 90^\circ$
4	90°	0	181	$q_4 + 180^\circ$
5	90°	0	0	$q_5 + 180^\circ$
6	90°	0	0	q_6

Tool: 60 mm along Z_6 .

How the table was obtained, and the trap in it. The joint axes were taken from the URDF at $q = 0$ and the common normals computed between consecutive axes. Every axis passes through the base Z line, so every a is zero except the perpendicular between the two parallel pitch axes. $d_4 = 181$, not 119: frame 3's origin sits where X_3 meets Z_3 ($z = 209$) and frame 4's where Z_4 meets Z_5 ($z = 390$), so the offset along Z_4 is $119 + 62$. Writing the "obvious" link length there puts the tool 193 mm out — which is exactly what the first version did.

2.3 Jacobian — velocity propagation, not differencing

$$\begin{aligned} {}^{i+1}_w \omega_{i+1} &= {}^{i+1}_i R \cdot {}^i_w \omega_i + \dot{\theta}_{i+1} \cdot \hat{Z} \\ {}^{i+1}_v v_{i+1} &= {}^{i+1}_i R \cdot ({}^i_v v_i + {}^i_w \omega_i \times {}^i_P_{i+1}) \end{aligned}$$

Propagated base to tip. This matters at the J4/J6 wrist singularity, where the two roll axes go collinear and a finite-difference Jacobian degrades.

2.4 Inverse kinematics

Bounded least-squares on position plus tool-axis direction. Two terms are non-obvious and both were arrived at by failure:

J6 is locked. A pen is rotationally symmetric about the tool axis, so J6 is pure redundancy. Free, the solve is under-determined and the solver wanders 135° between adjacent waypoints. Locking it makes an exactly-determined 5-DOF problem.

Null-space damping on J4 and J6 only. When $J5 \rightarrow 0$ those two roll axes become collinear and only their sum is determined. Damping all six joints equally suppressed the wander and destroyed accuracy — $0.02 \rightarrow 8.8$ mm rms — because it also fought the joints doing the task.

2.5 Statics

Gravity torque from a 60 000-pose sweep of the reachable configuration space, not a single worst-case pose:

joint	peak N·m	servo	stall N·m	margin
J1	0.00	ST3215	2.94	— (axis vertical)
J2	2.74	ST3250	4.90	1.79×
J3	1.39	ST3215	2.94	2.12×
J4	0.42	ST3215	2.94	7.0×
J5	0.42	ST3215	2.94	7.0×
J6	0.09	ST3215	2.94	32×

2.6 Compliance

Four closed-form terms, each stated so a disagreement is traceable: forearm bending (point + UDL), upper-arm bending, upper-arm slope × outboard length, and shell torsion from the payload's lateral offset.

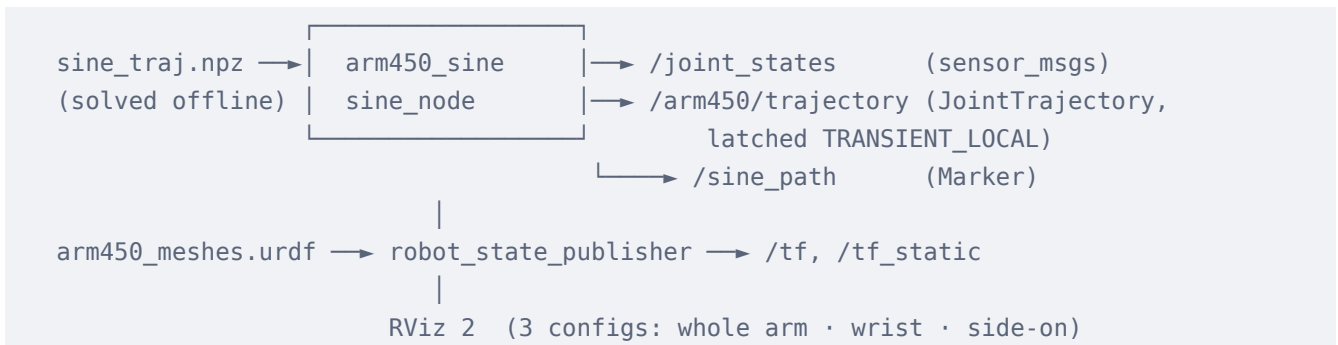
```
box section:  I = (b·h³ - (b-2t)(h-2t)³)/12
torsion:      J = 4·Am²·t / perimeter          (Bredt, closed thin tube)
cantilever:   δ = P·L³/(3EI)          UDL: δ = w·L⁴/(8EI)
```

3. Software architecture

```
cad/params.py    SINGLE SOURCE OF TRUTH – geometry, material, print settings
                  (MATERIAL = "PLA" drives E, ρ, yield, Tg, shrinkage)
├─ cad/*.py      CadQuery part models → STEP + STL
├─ audit_parts.py reads the STEP B-rep back; are the features PHYSICALLY there?
├─ dim_audit.py  every length, bore, fillet, chamfer, one at a time
├─ mate_check.py do holes in parts that bolt together actually line up?
├─ final_audit.py printability: walls, overhangs, build volume, orphan files
├─ fea.py        hex8 voxel FEA on the real printed geometry
├─ creep_fatigue.py Findley creep + Basquin S-N
├─ compliance_budget.py independent closed-form cross-check
├─ joint_loads.py gravity torque sweep, parameterised by tool reach
├─ verify_configs.py the whole load suite × 5 payload configurations
├─ dh_kinematics.py modified-DH FK + velocity-propagation Jacobian + IK
├─ check_ikpy.py  three-way cross-check against an outside package
├─ assemble.py   every part at its true mating position
├─ interference.py exact mesh-mesh collision (FCL)
├─ plan_sine.py  offline IK for the trajectory
└─ preflight.py  the 8-stage gate – one verdict
```

The design rule the whole codebase follows: every check reads the exported geometry, never the parameters used to author it. §4 explains why.

4. ROS 2 architecture



IK is solved offline and replayed. A least-squares solve per waypoint inside the node would block the executor. `plan_sine.py` writes 240 waypoints; the node replays them.

Three features exist because simulation exposed the need:

feature	what it prevents
cosine approach ramp	the node used to publish waypoint 0 on its first tick — invisible in RViz, a full-speed slam on real servos
ping-pong playback	modular wrapping is a discontinuity, not a loop: J4 jumps 43.3° in one 40 ms tick and the tool teleports 139.8 mm
latched JointTrajectory	/joint_states is a visualisation stream a controller will not follow

Packages: `arm450_description` (URDF, meshes, RViz), `arm450_sine` (node, launch). `Rsine` is a separate, frozen package — imported read-only for its plane geometry, never written to.

5. Simulation — parameters and results

5.1 Trajectory

parameter	value
board distance	300 mm (flange) / 342 mm (gripper tip)
trace centre height	220 mm
amplitude	40 mm
cycles	2
span	140 mm
waypoints	240
playback	25 Hz → 9.6 s per pass, ping-pong

5.2 Results

metric	value
position error, flange	0.014 mm rms, 0.217 max
position error, tool tip	0.055 mm rms, 0.834 max
tool tilt	0.093° rms
largest joint step between waypoints	2.09°
collisions, exact mesh	0 of 240
with gripper fitted	0 of 60
with docking probe fitted	0 of 60
peak joint rate	0.91 rad/s vs ~2.3 loaded ST3215

5.3 Workspace

Monte-Carlo FK, 200 000 poses inside the URDF joint limits, binned into a 5 mm (r, z) grid and revolved:

fitted	reach past J6	max reach	swept volume
bare face	—	450 mm	188 L
pen	30 mm	480 mm	239 L
gripper	42 mm	492 mm	262 L
dock probe	36 mm	486 mm	251 L

5.4 Cases that do not work

Horizontal-plate tracing fails. Swept 12 plate positions; best is 11.7 mm rms against 0.020 on the vertical board — 585× worse. J2 is hard against its +115° stop: to reach out over a table and tip the tool down, the shoulder runs out of travel. A 90° bent tool adapter was tried and improved it to 8.1 mm, at which point J2 and J5 both saturate.

It is a joint-range limit, not a solver or tool problem, and it traces to J5 being narrowed from $\pm 110^\circ$ to $-91^\circ/+49^\circ$ to stop the J4 housing striking the J6 body — a real collision found by exact-mesh sweep. The collision fix cost the horizontal capability, and that is a trade made knowingly.

6. Structural analysis

6.1 FEA

Hex8 voxel FEA on the actual printed geometry, with loads taken from the assembly-level statics rather than guessed.

method	part-wise submodelling, hex8 elements
loads	from joint_loads.py, not assumed

method	part-wise submodelling, hex8 elements
worst p99 von Mises safety factor	23.0 across all parts
working stress, worst case	0.437 MPa against 55 MPa yield

Why part-wise and not assembly-level: an assembly FEA needs joint stiffness, bolt preload, and bearing radial/moment stiffness — none of which is known to better than a factor of two before hardware. Part-wise uses loads that are known and returns each part's margin in isolation. This is a stated limitation, not an oversight (§10.2).

6.2 Creep

Findley power law, 1 year of continuous load:

working stress	0.437 MPa
stress ratio	0.0064 of yield
strain at 1 year	0.097 % (plain PLA)
limit	0.5 %

The absolute strain is tiny but it is several times the elastic strain, and the Findley fit is being used two orders of magnitude below its calibration range. Actionable outcome: park the arm folded when idle, not extended.

6.3 Fatigue

Basquin S-N. No endurance limit is assumed — polymers do not have one, so a life target is used instead of a fatigue limit.

stress amplitude	0.22 MPa
cycles to failure	8.0×10^{31}
target	1×10^8

Not limiting, because the arm is stiffness-driven, not strength-driven.

6.4 Stiffness and the precision budget

config	structural	0.5° backlash	combined
bare	0.071 mm	1.88 mm	1.88 mm
gripper	0.108 mm	2.24 mm	2.24 mm
gripper + 225 g	0.197 mm	2.24 mm	2.25 mm

Backlash exceeds structural deflection by roughly 20× in every configuration. This arm is servo-limited, not structure-limited. No change to the printed parts moves the number. Accuracy work belongs in joint-side feedback.

6.5 Mass

group	mass
printed structure (PLA, 90 % effective fill)	599 g
servos, 6 ×	360 g
bearings, 12 ×	176 g
aluminium shafts, 6 ×	205 g
structure subtotal	1340 g
fasteners and heat-set inserts	244 g
ASSEMBLED	1584 g
budget	1450 g
over by	134 g

This is an open item and are not hiding it. See §9 Q11.

7. Verification method — the part are most confident about

7.1 The principle

Every check reads the exported STEP or mesh, never the parameters used to author it. An early gate checked BEARING_POCKET_D == 42.0 and passed 55/55 while three bearing pockets did not physically exist: a hollowing operation had removed the material the pocket was cut into, and subtracting a Ø42 pocket from an already-empty region succeeds silently and changes nothing.

A boolean that removes nothing does not fail. It returns the same solid, and every step downstream — render, drawing, mass, collision — treats the result as correct.

7.2 The eight-stage gate

stage	question	criterion	result
1 Design check	do the features physically exist?	STEP B-rep audit	52 features, 0 missing
2 FEA	does anything yield?	p99 SF $\geq$ 3	23.0
2 Load cases	with and without a tool?	6 criteria × 5 configs	30/30
3 Creep + fatigue	does it fail slowly?	< 0.5 %/y, > 1e8	pass
4 Stiffness	does the tool stay put?	$\leq$ 0.30 mm	0.197
5 URDF	does the model match the machine?	parses, mass, meshes	pass
6 Path tracing	does the real trajectory work?	0 collisions	240/240

stage	question	criterion	result
7 Simulation	will ROS work on hardware?	ramp, topic, rate	pass
8 End effectors	do the tools fit, does it latch?	32 checks + a mate	pass

Current: 25 passed, 1 failure — the mass overage in §6.5.

7.3 Independent cross-checks

Nothing important rests on one calculation:

quantity	primary	cross-check	agreement
forward kinematics	Craig DH	URDF chain and ikpy	2×10^{-13} mm
Jacobian	velocity propagation	finite difference	1.8×10^{-5}
compliance	verify_configs	compliance_budget.py	0.098 mm both (CF case)
mass	STEP volumes × density	per-part geometry	—
bearing fit	parametric	virtual press into every pocket	light press, 10/10

Three implementations sharing no code cannot agree on the same mistake. That is why ikpy is in the loop: agreement between our DH table and our URDF would only prove the table copies the URDF faithfully, errors included.

7.4 What the method returns

The value of reading exported geometry rather than authoring parameters is not theoretical. Fifteen defects were identified by this route that had survived visual inspection of the same parts, and they fall into four recurring classes:

Operations that succeed without effect. A boolean subtraction from a region that is already void returns the original solid unchanged, with no error. Three bearing pockets were absent for this reason while a parameter check reported every dimension correct.

Assemblies that are not connected. A union of two bodies that do not touch also succeeds; it simply returns a compound. The docking probe exported as four separate solids, three of them latch lugs suspended clear of the bore they were intended to grow from.

Dimensions never checked against what they cut. A capture groove specified at 3.30 mm depth was applied to a wall 1.65 mm thick, severing the feature it was cut into.

Relationships that appear in no single-part view. The capture cone was specified deeper than the mating spigot was long, so probe and target could not engage at any insertion depth or roll angle. Neither part is incorrect in isolation; the defect exists only in the pair.

Two further categories were found in the documentation rather than the geometry: drawings asserting features the parts did not carry, and constants duplicated across files

that had since diverged.

The common property is that none of these failed loudly. Each produced a plausible artefact — a render, a mass figure, a passing check — that was wrong. Detection required interrogating the exported representation rather than the intent behind it.

8. Space vs ground manipulators

8.1 The differences that matter

	ground manipulator	space manipulator
Gravity	dominates sizing; the arm must hold itself up	~0 g. Sizing is driven by inertia and dynamics, not weight
What limits payload	actuator torque against gravity	reaction on the base — the arm moves the spacecraft
Base	bolted to a rigid floor	free-floating. Moving the arm rotates the satellite; momentum must be managed
Structure	stiff enough not to sag	stiff enough not to oscillate — no damping in vacuum
Thermal	20 ± 15 °C	–120 to +120 °C cycling; CTE mismatch drives joint clearances
Lubrication	wet grease	dry film (MoS ₂) or PFPE — wet grease outgasses and creeps
Materials	anything	TML < 1 %, CVCM < 0.1 %; no trapped volumes
Radiation	none	TID-rated electronics, latch-up immunity
Backlash	tolerable	often unacceptable — no operator to correct in the loop
Repair	trivial	none. Single-fault tolerance or redundancy
Qualification	"it works"	vibration, shock, thermal-vac, EMC, life test
Control	position	often impedance/force — contact dynamics with a free target

8.2 The one that surprises people

A space arm's hardest problem is not holding the load — it is that there is nothing to push against. On the ground the floor absorbs every reaction. In orbit, Newton's third

law means accelerating the arm rotates the spacecraft. Canadarm2 moves slowly for this reason, and free-flying servicers must plan trajectories that are reactionless or compensate with reaction wheels.

ARM-450 cannot demonstrate this, because it is bolted down. That is the honest limit of the analogy, and MASCOT's maglev float is precisely the attempt to recover it.

8.3 How ARM-450 relates

Of sixteen subsystems compared against a flight equivalent: 3 survive unchanged, 2 need rework, 11 must be replaced.

Survives:

- The kinematics. A 6R chain with this layout is unaffected by what it is made of. URDF, IK and workspace transfer directly.
- The joint architecture. One bore, one shaft, one bearing family is good practice at any grade; only part numbers and lubricant change.
- The fastening scheme. Bolt patterns and load paths transfer; flight versions get locking compound, staking and vented screws in the same places.

Does not:

- The material is the wall. PLA's glass transition is $\sim 60^\circ\text{C}$; a sun-facing surface in LEO exceeds it. It outgasses. Re-made in Al 6061-T6 the same geometry is $\approx 1.13\text{ kg}$ — lighter than printed — and about $10\times$ stiffer. The swap is an improvement that costs money, not a compromise.
- The actuators. Hobby serial servos: plastic gears, wet grease, no radiation tolerance, no joint-side feedback. This single choice sets the 2.2 mm precision figure.
- Vacuum compatibility was never designed in. The printed parts contain closed internal volumes with no vent path.
- Nothing has been qualified.

The honest framing: ARM-450 is a kinematic and architectural prototype of a space manipulator, and a functional demonstrator of a docking interface. It is not a flight article. What transfers is the geometry, the joint scheme, the docking principle — and the verification method, which is grade-independent.

8.4 What genuinely is space-relevant

The docking interface. Capture tolerance was solved as geometry, not accuracy: a $\varnothing 34$ mouth over a $\varnothing 16$ throat gives a measured $\pm 9.75\text{ mm}$ envelope against $\approx 2.2\text{ mm}$ of arm error. This is exactly how real docking hardware works, and it is the transferable idea.

Fluid transfer. A radial O-ring in the spigot, compressed by the bayonet itself, with a $\varnothing 8$ pass-through in both halves — the architecture of an orbital refuelling coupling at demonstration scale.

No seventh actuator. Using an existing DOF to drive a mechanism is a mass-saving technique that matters far more in orbit than on a bench.

9. Anticipated questions

Written to be answered honestly. Where the answer is "this does not know", it says so.

Q1. Has the arm been built? What has actually been verified?

No. Nothing is printed. Everything is analytical or simulated, and §10 lists the seven measurements outstanding with their predictions and falsification conditions. What is verified is that the geometry is self-consistent, the loads are within margin, the trajectory is collision-free on exact meshes, and the kinematics agrees with two independent implementations. That is design verification, not hardware validation, and we do not claim otherwise.

Q2. A safety factor of 23 — is the arm massively over-designed?

Yes on strength, and deliberately. The arm is stiffness-driven, not strength-driven — sized by how far the tool deflects, not by whether anything breaks. Once the wall is thick enough to be stiff, it is far more than thick enough to be strong. Reporting SF 23 is reporting that strength was never the binding constraint. The binding constraint is backlash, which beats structural deflection by 20×.

Q3. Then why not thin the walls and save mass?

Because bending stiffness goes as t^3 . Dropping the wall from 2.4 to 1.8 mm — a 25 % reduction — costs 58 % of the stiffness. And at a 0.6 mm nozzle, 2.4 mm is exactly four perimeters: thinner walls stop being solid extrusion and start containing infill voids, which is a different material, not a thinner one.

Q4. Why plain PLA rather than an engineering polymer?

Availability and print reliability. The full load suite was re-run for it — 10 of 10 pass, worst deflection 0.197 mm against a 0.30 mm limit — and at the tool the difference from carbon-filled is 2.24 vs 2.25 mm, because backlash dominates. PLA is not a compromise for this demonstrator. It would be for a flight article, and §8.3 says so.

Q5. 2.2 mm positional accuracy is poor for a manipulator.

It is, and it is honest. It is set almost entirely by servo backlash — the printed structure contributes 0.197 mm, about 2 % of the total. The arm uses hobby serial servos with no joint-side encoder. Improving it means adding feedback at the joint, not stiffening the arm; a magnetic encoder on each output shaft would take it to ~0.1 mm.

And it is why the docking interface has a ± 9.75 mm capture cone. designed the interface around the accuracy have rather than pretending to accuracy this does not.

Q6. Simulation reports 0.055 mm rms. Is that the claimed accuracy?

No. That number validates the kinematics — the solver puts the tool where it is asked. The hardware prediction is 1–3 mm rms, recorded in §10 with its falsification condition. If the trace measures under 0.5 mm, our backlash model is wrong and will say so.

Q7. Why two bearings per joint? Is that not wasteful?

A single deep-groove bearing is a pivot — it permits tilt. Two spaced apart resist bending. The measured cost of removing one: tool error goes 0.45 mm → 28.90 mm, 64× worse. An earlier revision of this arm made exactly that mistake with a single thrust washer, and it is documented as the largest error in the project.

Q8. Why is the wrist over-engineered with 42 mm bearings?

It is not, any more — that was measured and fixed. The wrist runs 6706 (Ø37 × 4) against 6806 (Ø42 × 7) at the shoulder: 24 g saved, no stiffness lost, because moment stiffness goes as bearing spacing squared and is independent of width. What remains is the cost of commonality: one bore diameter across all six joints, so one fit characterisation and one spare part number.

Q9. Why is the J5 range asymmetric, -91° to $+49^\circ$?

Because an exact-mesh sweep found the J4 housing striking the J6 output body over the excluded range. It was specified as $\pm 110^\circ$; the collision check narrowed it, and the URDF carries the narrowed limit while the collision checker parses limits from the URDF rather than holding a second copy.

It cost us horizontal-plate tracing (\$5.4). The trade was made knowingly: an arm that cannot trace a table is better than an arm that destroys its own wrist.

Q10. Can it trace on a table?

No. Best of 12 plate positions is 11.7 mm rms against 0.020 on a vertical board. J2 saturates at $+115^\circ$. A 90° bent tool adapter was tried; then J2 and J5 both saturate. It is a joint-range limit. The workaround costing nothing is to mount the arm on its side, which makes the table a vertical board in the arm's own frame.

Q11. The mass budget is 1450 g and the arm is 1584 g. Explain.

The original figure counted structure, actuators, bearings and shafts. It had no fastener term — 40 brass heat-set inserts, six M5 × 90 preload bolts and roughly a hundred screws, 244 g in total, 15 % of the arm and entirely absent. It was identified, quantified every fastener from its own geometry, and put it in the gate as a hard failure, which is why the gate currently reads NO-GO.

Available savings were costed: aluminium shims (−19 g), M4 × 80 preload bolts instead of M5 × 90 (−41 g), titanium seam screws (−15 g), nylon washers (−3 g) — 78 g, leaving ~56 g still over. Closing it entirely requires lightening the base, which invalidates its FEA until re-run.

The overage is reported, not absorbed, and the budget has not been quietly raised.

Q12. What if the bearing does not fit the printed pocket?

That is the single largest risk and it has a dedicated test. The fit coupon carries three pockets at Ø41.95 / 42.00 / 42.05, three insert-boss sizes and a seam sample; it prints in 0.9 h and gates the ~19 h of parts that contain a bearing pocket. Whichever diameter gives a firm thumb press sets BRG_FIT, everything re-exports, and the structural parts print.

Plain PLA shrinks 0.40 % — 0.17 mm on a Ø42 bore, larger than the coupon's whole 0.10 mm bracket — so all three may come out tight. That is the coupon working: set XY compensation ≈ -0.15 mm and run it again. Two coupon prints are budgeted two coupon prints.

Q13. On what basis is the FEA considered correct?

It is not relied on alone. It is cross-checked against a closed-form compliance budget

written from first principles with every formula stated, and the two agree. The FEA is also deliberately part-wise, not assembly-level, because assembly FEA needs joint stiffness and bolt preload that are not known before hardware. §10.2 lists this as a limitation.

Q14. Why offline IK? Is that not a limitation?

For a repeated demonstration trajectory it is the correct choice: a least-squares solve per waypoint inside the ROS node would block the executor, and the path is identical every run. The node also publishes the solved path as a latched JointTrajectory, so a real controller can consume it directly. For reactive tasks — following a moving target — this would need to become online IK, and that is named in the future scope.

Q15. What happens if a servo fails mid-trace?

The arm is single-string — there is no redundancy, and it would stop. On a bench that is acceptable. §8.1 lists redundancy as one of the eleven subsystems that must be replaced for flight.

Q16. Is this actually space-relevant, or is it a robot arm with a story attached?

Three things transfer and each is defensible: the kinematics and joint architecture (unaffected by material), the docking principle (capture tolerance solved as geometry, which is how real docking hardware works), and the verification method (grade-independent). Eleven of sixteen subsystems do not transfer, and §8.3 names every one.

It is preferable to state that than claim more.

Q17. What is the weakest element of the work?

That backlash is assumed and not measured, and it dominates the error budget. The headline precision figure is therefore the least evidenced number in the document. It is first in the measurement plan.

Second: the fit coupon is unprinted, so this does not yet know what this printer does to a Ø42 bore.

Q18. What would be done differently from the outset?

Two things. Interrogate the geometry from day one — the parameter-checking gate passed 55/55 while three bearing pockets did not exist, and everything since has been reading the exported STEP instead. And put every material and print constant in one place from the start: E, density and yield were hardcoded in twelve files, so switching material meant finding all twelve, and missing one would have left the gate silently mixing two materials.

10. Measurement plan

No part of this design has yet been built, so every figure in the preceding sections is analytical or simulated. The measurements below are the ones that convert those figures into evidence, and they are listed here in the form they will be executed.

Each carries a prediction and a falsification condition, both fixed in writing before the measurement is taken. The prediction states what the analysis expects; the falsification condition states the result that would show the analysis to be wrong. A test that cannot

fail confirms nothing, so a measurement with no falsification condition is not recorded as a test. The final column is left blank and is completed when the measurement is made.

10.1 Planned measurements

#	measurement	method	prediction	falsified if	result
1	Ø42 bore fit	fit coupon, thumb press	Ø42.00 correct, or -0.15 XY comp	drops in free / will not start	_____
2	servo backlash	dial indicator at the tool	0.3–1.0° per joint, 1.9–2.2 mm at tip	< 0.5 mm or > 5 mm	_____
3	sine trace accuracy	trace on paper, scan, fit	1–3 mm rms	< 0.5 mm — backlash model wrong	_____
4	docking latch	bench, by hand, roll J6 30°	latches and releases repeatably	lugs bind or release under load	_____
5	static deflection	300 g at full extension	0.197 mm	> 0.30 mm — seam not closing	_____
6	J2 torque and case temp	bus current + IR after 3 min	2.74 N·m, 44 °C	stalls, or case > 60 °C	_____
7	assembled mass	scale	1584 ± 50 g	outside 1534–1634 g — volume or density model wrong	_____

10.2 Stated limitations

1. No hardware. Every number is analytical or simulated.
2. FEA is part-wise, not assembly-level — joint stiffness, bolt preload and contact are not modelled.
3. Backlash is assumed, not measured, and it dominates the error budget.
4. Servo case-screw positions are from a datasheet, not a measured servo.
5. Creep uses a Findley fit two orders of magnitude below its calibration range.
6. Print anisotropy is not modelled — layer adhesion is typically 40–70 % of in-plane strength; the FEA is isotropic.
7. The docking mate is verified as rigid-body geometry — no friction, compliance or misalignment dynamics.
8. The fluid line is not in any collision model; internal routing through the joints was

measured as impossible (Ø3.6 mm free channel against a Ø6 tube).

11. Future scope

11.1 Immediate — weeks

Print and measure	close all seven items in §10. Until then this is a design, not a machine.
Joint-side encoders	the single highest-value change. Magnetic encoders on each output shaft would take precision from ~2.2 mm to ~0.1 mm — a 20× improvement for roughly 30 g and six I ² C devices, and it addresses the one thing no structural change can.
Fluid transfer	the sealed target is designed and verified; the tube route is not. Internal routing is ruled out by measurement; external needs a designed path.

11.2 Medium — months

Online IK for reactive tasks — following a moving target rather than replaying a solved path. Requires the solve to leave the ROS node's executor thread.

Wrist redesign to recover J5 travel and with it horizontal-plate tracing. The constraint is the J4-housing/J6-body collision; a re-proportioned yoke would buy back the range.

Force/impedance control. Docking with a free-floating target is a contact-dynamics problem, not a position problem. This is the single biggest control-side gap between this arm and a real servicer.

Aluminium version. Same geometry in Al 6061-T6: ≈1.13 kg, ~10× stiffer, no glass transition at 60 °C. It also removes the mass overage entirely.

11.3 Long — the MASCOT path

Reactionless trajectory planning. Once the base floats, arm motion rotates the platform. Planning motions with zero net angular momentum — or compensating with reaction wheels — is the defining problem of free-flying manipulation, and it cannot be studied on a bolted-down arm. The maglev platform is what makes it accessible.

Full refuelling sequence: approach → capture in the cone → latch by J6 → seal → transfer → purge → unlatch → retreat, closed-loop and autonomous.

Vacuum-compatible revision: vented parts, dry-film bearing lubricant, outgassing-qualified materials. This is a redesign, not a modification, and §8.3 says which eleven subsystems it touches.

12. Supporting documents

document	contents
ARM450_PAPER.pdf	the work as a research paper
ARM450_REPORT.pdf	full engineering report — FEA, creep, fatigue, stiffness
ARM450_KINEMATICS.pdf	DH derivation and the three-way cross-check
ARM450_PREFLIGHT.pdf	the eight-stage gate
ARM450_DRAWINGS_ISO.pdf / _3VIEW.pdf	dimensioned sheets, 19 parts
ARM450_BUY.pdf	procurement and print order
ARM450_SLICING.pdf	print settings, and the two that are not preferences
ARM450_END_EFFECTORS.pdf	gripper and docking probe
ARM450_WORKSPACE.pdf	reach and swept volume
ARM450_LESSONS.pdf	the mistakes and the rules they produced
ARM450_LOG.pdf	dated record of every change and why
ARM450_COMMANDS.pdf	every command, generated from the source

Reproduce everything:

```
python3 preflight.py      # the 8-stage gate, one verdict
python3 check_ikpy.py    # kinematics, three ways
python3 verify_configs.py # load suite × 5 payload configurations
python3 dim_audit.py     # every dimension, one at a time
python3 mate_check.py    # do mating parts line up
python3 final_audit.py   # printability
```